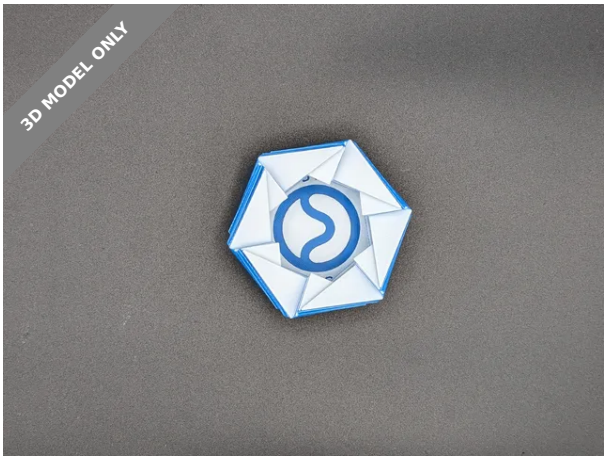

Origami Bloom Pattern Foldable

BYU CMR

VIEW IN BROWSER

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Summary

This model is an example of a Yoshimura bloom pattern, with sixfold symmetry and two rings of folds.

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Summary

This is a 3D-printable Yoshimura bloom pattern, specifically a Y-6.2 pattern (meaning it has six-fold symmetry and two rings extend out from the center). The membranes between the panels are designed to accommodate the folding process, and allow the model to rest in both the closed and the open states.

See this YouTube video by EzOrigami for instructions on how to fold your own:

BYU also made a video discussing the details of these patterns and their potential in engineering mechanisms and beyond:

Printing Instructions

This model is best printed in PETG, or some other similarly flexible material (PLA also works fine). **The model, at its current scaling, is intended to be printed with two 0.1mm layers as the base.** This ensures that all folds have two overlapping layers of filament on which to bend (printing with a single 0.2mm thick bottom layer will mean that some of the hinges may break very easily). Here is a tutorial on how to modify individual layer heights in PrusaSlicer.

You're welcome to experiment with printing the first two layers out of a more flexible material, and then printing the remaining layers with something stiffer. PLA worked fine for the models we made on our own printers. See this post for a tutorial on how to make your own living hinges.

It's best to remove your print from the build plate sooner rather than later: the residual heat from the plate will help you bend the creases into the model (see this video for which directions to bend them in).

You're welcome to experiment with printing the first two layers out of a more flexible material, and then printing the remaining layers with something stiffer. PLA worked fine for the demos we made.

Learn More

This design was developed by the Compliant Mechanisms and Robotics Group (CMR) from Brigham Young University (BYU). Follow us at @byucmr on Instagram, @CompliantMechanismsResearchGroup on Facebook, or visit the BYU Compliant Mechanisms and Robotics (CMR) website to learn more about compliant mechanisms.

Technical Information

To read the backround research behind this work, visit: https://doi.org/10.1098/rspa.2025.0299

Supplemental material can be found at ScholarsArchive: https://scholarsarchive.byu.edu/data/85/

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Model files

y6-2-printable.stl

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